

Matrix Rank Solutions Using Elementary Row Operations

Problem 1: Matrix $[2 \ 0 \ -1; 5 \ 1 \ 0; 0 \ 1 \ 3]$

Step 1: Write the original matrix

$$\begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$$

Step 2: $R_2 \rightarrow R_2 - (5/2)R_1$

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 1 & 5/2 \\ 0 & 1 & 3 \end{bmatrix}$$

Step 3: $R_3 \rightarrow R_3 - R_2$

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 1 & 5/2 \\ 0 & 0 & 1/2 \end{bmatrix}$$

Step 4: $R_3 \rightarrow 2R_3$

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 1 & 5/2 \\ 0 & 0 & 1 \end{bmatrix}$$

Result: The matrix is in row echelon form with 3 non-zero rows. **Rank = 3**

Problem 2: Matrix $[1 \ -2 \ 3; 2 \ 1 \ 2; 5 \ -5 \ 11]$

Step 1: Write the original matrix

$$\begin{bmatrix} 1 & -2 & 3 \\ 2 & 1 & 2 \\ 5 & -5 & 11 \end{bmatrix}$$

Step 2: $R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 5R_1$

[1 -2 3]
[0 5 -4]
[0 5 -4]

Step 3: $R_3 \rightarrow R_3 - R_2$

[1 -2 3]
[0 5 -4]
[0 0 0]

Result: The matrix has 2 non-zero rows. **Rank = 2**

Problem 3: Matrix [2 1 -2; -1 -1 1; 3 1 -2]

Step 1: Write the original matrix

[2 1 -2]
[-1 -1 1]
[3 1 -2]

Step 2: $R_1 \leftrightarrow R_2$ (swap rows to get leading 1)

[-1 -1 1]
[2 1 -2]
[3 1 -2]

Step 3: $R_1 \rightarrow -R_1$

[1 1 -1]
[2 1 -2]
[3 1 -2]

Step 4: $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - 3R_1$

[1 1 -1]
[0 -1 0]
[0 -2 1]

Step 5: $R_2 \rightarrow -R_2$

[1 1 -1]
[0 1 0]
[0 -2 1]

Step 6: $R_3 \rightarrow R_3 + 2R_2$

[1 1 -1]
[0 1 0]
[0 0 1]

Result: The matrix has 3 non-zero rows. **Rank = 3**

Problem 4: Matrix [1 2 -1 1; 2 3 4 5; 1 4 -13 -5]

Step 1: Write the original matrix

[1 2 -1 1]
[2 3 4 5]
[1 4 -13 -5]

Step 2: $R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - R_1$

[1 2 -1 1]
[0 -1 6 3]
[0 2 -12 -6]

Step 3: $R_2 \rightarrow -R_2$

[1 2 -1 1]
[0 1 -6 -3]
[0 2 -12 -6]

Step 4: $R_3 \rightarrow R_3 - 2R_2$

[1 2 -1 1]
[0 1 -6 -3]
[0 0 0 0]

Result: The matrix has 2 non-zero rows. **Rank = 2**

Problem 5: Matrix [1 -2 1 -1; 1 1 -2 3; 4 1 -5 8]

Step 1: Write the original matrix

```
[1 -2 1 -1]
[1 1 -2 3]
[4 1 -5 8]
```

Step 2: $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - 4R_1$

```
[1 -2 1 -1]
[0 3 -3 4]
[0 9 -9 12]
```

Step 3: $R_3 \rightarrow R_3 - 3R_2$

```
[1 -2 1 -1]
[0 3 -3 4]
[0 0 0 0]
```

Result: The matrix has 2 non-zero rows. **Rank = 2**

Problem 6: Matrix [1 2 1; 2 3 4; 1 3 -1; 8 13 14]

Step 1: Write the original matrix

```
[1 2 1]
[2 3 4]
[1 3 -1]
[8 13 14]
```

Step 2: $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - R_1$, $R_4 \rightarrow R_4 - 8R_1$

```
[1 2 1]
[0 -1 2]
[0 1 -2]
[0 -3 6]
```

Step 3: $R_2 \rightarrow -R_2$

[1 2 1]
[0 1 -2]
[0 1 -2]
[0 -3 6]

Step 4: $R_3 \rightarrow R_3 - R_2$, $R_4 \rightarrow R_4 + 3R_2$

[1 2 1]
[0 1 -2]
[0 0 0]
[0 0 0]

Result: The matrix has 2 non-zero rows. **Rank = 2**

Problem 7: Matrix [1 2 3 4; 2 3 4 5; 7 11 15 19; 9 15 21 27]

Step 1: Write the original matrix

[1 2 3 4]
[2 3 4 5]
[7 11 15 19]
[9 15 21 27]

Step 2: $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - 7R_1$, $R_4 \rightarrow R_4 - 9R_1$

[1 2 3 4]
[0 -1 -2 -3]
[0 -3 -6 -9]
[0 -3 -6 -9]

Step 3: $R_2 \rightarrow -R_2$

[1 2 3 4]
[0 1 2 3]
[0 -3 -6 -9]
[0 -3 -6 -9]

Step 4: $R_3 \rightarrow R_3 + 3R_2$, $R_4 \rightarrow R_4 + 3R_2$

```
[1 2 3 4]
[0 1 2 3]
[0 0 0 0]
[0 0 0 0]
```

Result: The matrix has 2 non-zero rows. **Rank = 2**

Problem 8: Matrix [1 1 1 1; -1 1 1 -1; 1 0 1 1; 1 1 0 1]

Step 1: Write the original matrix

```
[1 1 1 1]
[-1 1 1 -1]
[1 0 1 1]
[1 1 0 1]
```

Step 2: $R_2 \rightarrow R_2 + R_1$, $R_3 \rightarrow R_3 - R_1$, $R_4 \rightarrow R_4 - R_1$

```
[1 1 1 1]
[0 2 2 0]
[0 -1 0 0]
[0 0 -1 0]
```

Step 3: $R_2 \rightarrow R_2/2$

```
[1 1 1 1]
[0 1 1 0]
[0 -1 0 0]
[0 0 -1 0]
```

Step 4: $R_3 \rightarrow R_3 + R_2$

```
[1 1 1 1]
[0 1 1 0]
[0 0 1 0]
[0 0 -1 0]
```

Step 5: $R_4 \rightarrow R_4 + R_3$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Result: The matrix has 3 non-zero rows. **Rank = 3**

Problem 9: Matrix $\begin{bmatrix} 2 & 0 & -1 & 0 \\ 4 & 1 & 0 & 5 \\ 0 & 1 & 3 & 6 \\ 6 & 1 & -2 & 6 \end{bmatrix}$

Step 1: Write the original matrix

$$\begin{bmatrix} 2 & 0 & -1 & 0 \\ 4 & 1 & 0 & 5 \\ 0 & 1 & 3 & 6 \\ 6 & 1 & -2 & 6 \end{bmatrix}$$

Step 2: $R_1 \rightarrow R_1/2$

$$\begin{bmatrix} 1 & 0 & -1/2 & 0 \\ 4 & 1 & 0 & 5 \\ 0 & 1 & 3 & 6 \\ 6 & 1 & -2 & 6 \end{bmatrix}$$

Step 3: $R_2 \rightarrow R_2 - 4R_1$, $R_4 \rightarrow R_4 - 6R_1$

$$\begin{bmatrix} 1 & 0 & -1/2 & 0 \\ 0 & 1 & 2 & 5 \\ 0 & 1 & 3 & 6 \\ 0 & 1 & 1 & 6 \end{bmatrix}$$

Step 4: $R_3 \rightarrow R_3 - R_2$, $R_4 \rightarrow R_4 - R_2$

$$\begin{bmatrix} 1 & 0 & -1/2 & 0 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

Step 5: $R_4 \rightarrow R_4 + R_3$

[1 0 -1/2 0]
[0 1 2 5]
[0 0 1 1]
[0 0 0 2]

Result: The matrix has 4 non-zero rows. **Rank = 4**

Problem 10: Matrix [2 3 1 0 4; 3 1 2 -1 1; 4 -1 3 -2 -2; 5 4 3 1 3]

Step 1: Write the original matrix

[2 3 1 0 4]
[3 1 2 -1 1]
[4 -1 3 -2 -2]
[5 4 3 1 3]

Step 2: $R_1 \rightarrow R_1/2$

[1 3/2 1/2 0 2]
[3 1 2 -1 1]
[4 -1 3 -2 -2]
[5 4 3 1 3]

Step 3: $R_2 \rightarrow R_2 - 3R_1$, $R_3 \rightarrow R_3 - 4R_1$, $R_4 \rightarrow R_4 - 5R_1$

[1 3/2 1/2 0 2]
[0 -7/2 1/2 -1 -5]
[0 -7 1 -2 -10]
[0 -7/2 1/2 1 -7]

Step 4: $R_2 \rightarrow -2R_2/7$

[1 3/2 1/2 0 2]
[0 1 -1/7 2/7 10/7]
[0 -7 1 -2 -10]
[0 -7/2 1/2 1 -7]

Step 5: $R_3 \rightarrow R_3 + 7R_2$, $R_4 \rightarrow R_4 + (7/2)R_2$


```
[1  3/2  1/2  0  2]
[0  1  -1/7  2/7  10/7]
[0  0  0  0  0]
[0  0  0  2  0]
```

Step 6: Swap R_3 and R_4

```
[1  3/2  1/2  0  2]
[0  1  -1/7  2/7  10/7]
[0  0  0  2  0]
[0  0  0  0  0]
```

Result: The matrix has 3 non-zero rows. **Rank = 3**

Summary of Results:

1. Rank = 3
2. Rank = 2
3. Rank = 3
4. Rank = 2
5. Rank = 2
6. Rank = 2
7. Rank = 2
8. Rank = 3
9. Rank = 4
10. Rank = 3